Yogesh Venkataramanan

Dearborn, MI | 313-367-8506 | yogeshvenkat21@gmail.com | linkedin.com/in/yogeshvenkat21 | github.com/Yogesh21v | yogesh21v.github.io

Experience

NowforceAI Jul 2026 – Present
Junior Software Engineer *Aubrey, TX (Remote)*

- Designing and building a full-stack, multi-tenant billing and finance module (parties, items, sales, purchases, cash & bank, salary tracking, and financial reporting) for a production applicant-tracking system, using Node.js/Express and MySQL on the backend and vanilla JavaScript on the frontend.
- Diagnosing and resolving database compatibility issues in production API routes as the module scales, including a defect where UPDATE statements were silently failing, to keep edit functionality reliable across the billing system.
- Architecting an evolving salary and resource-tracking system to manage incoming client payments and outgoing consultant payouts for bench and placed staff, including per-resource expense tracking and financial reports (P&L, day-book, party statements).
- Owning the feature end-to-end – database schema, REST API design, frontend UI, and production deployment – iterating based on business requirements and shipping updates through a backup-and-verify release process.

Imagine Marketing Services Pvt. Ltd. (boAt) Jan 2024 – May 2024
R&D Intern *Bangalore, India*

- Developed a scalable, Python-based framework to assess the State of Health (SoH) of batteries in True Wireless Stereo (TWS) devices using voltage, temperature, and usage telemetry, enhancing longevity prediction and performance analysis.
- Built an intuitive Flask-based web interface for dynamic user input and integrated it with an Arduino WeMOS D1R1 UNO for real-time data acquisition, visualization, and diagnostic analysis.
- Implemented an ML-based classification model that categorized battery SoH as “Normal” or “Low,” enabling proactive battery management, supporting proactive diagnostics, and reducing power inefficiency.

Projects

ShaktiConnect | *Full-Stack Community Web Platform* May 2026 – Aug 2026
Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, JavaScript, REST APIs, Playwright

- Built a full-stack web platform from scratch with a custom REST API (Node.js/Express), a MongoDB database, and JWT-based authentication, powering a community forum, member directory, and resource-center features.
- Implemented core functionality including live voting, threaded replies, search/filtering, and secure user sessions, then validated all workflows through automated end-to-end testing (Playwright) before deployment.
- Designed a cost-efficient architecture using a self-provisioning local database and a free-tier deployment path, prioritizing scalability without added infrastructure cost.

Soccer Concierge Chatbot | *AWS Serverless* Sep 2025 – Dec 2025
AWS Lambda, API Gateway, DynamoDB, SNS/SQS, CloudWatch, REST APIs · github.com/Yogesh21v/goalpost-chatbot

- Designed a serverless chatbot using AWS Lambda and API Gateway to process user queries and return real-time soccer scores and fixtures.
- Implemented intent-based routing, entity normalization, fallback handling, and third-party REST API integration.
- Logged conversations in DynamoDB and enabled asynchronous workflows using SNS/SQS with monitoring via CloudWatch.

SmartDoor | *Face Recognition Access System (AWS)* Sep 2025 – Dec 2025
AWS Rekognition, Kinesis Video Streams, Lambda, DynamoDB, IAM, SNS · github.com/Yogesh21v/smartdoor

- Built a serverless access-control system using AWS Rekognition and Kinesis Video Streams for real-time visitor detection.
- Implemented OTP-based authentication workflows using Lambda, DynamoDB (TTL), and SNS notifications.
- Secured APIs and services using IAM roles and least-privilege access control.

News Article Retrieval Engine | *Bayesian Network* Jan 2025 – Apr 2025
Python, Bayesian Inference, Information Retrieval, NLP · github.com/Yogesh21v/news-retrieval-system

- Built a probabilistic news search engine using a Bayesian Network model to retrieve relevant news articles from a corpus of 210k+ HuffPost headlines, modeling relationships between terms, documents, and categories.
- Developed the query processing, Bayesian inference mechanism, and ranking algorithm, enabling efficient and accurate document retrieval based on user input.

- Implemented evaluation metrics such as precision, recall, and Mean Average Precision (MAP) to assess the retrieval effectiveness of the system.

Lip Reading AI | *LipNet Architecture* Jan 2025 – Apr 2025

TensorFlow, Keras, OpenCV, Deep Learning, Computer Vision · github.com/Yogesh21v/lipnet-lipreading-ai

- Designed and implemented a LipNet-based deep learning model combining 3D CNNs, Bi-Directional LSTMs, and CTC loss to perform visual-only speech recognition from silent video clips.
- Built a complete training pipeline using TensorFlow, Keras, and OpenCV, handling data preprocessing (frame extraction, normalization), training (batching, checkpointing), and evaluation.
- Trained the model on the GRID dataset, achieving real-time character-level predictions from lip movements, with potential applications in accessibility, silent surveillance, and AI assistants.

IoT-Based Railway Track Crack Detection System Jun 2023 – Nov 2023

Arduino UNO, GSM, GPS, Ultrasonic Sensors

- Engineered a GPS-enabled autonomous vehicle using Arduino UNO and ultrasonic sensors to detect cracks and send real-time SMS alerts with Google Maps links.
- Integrated GSM and GPS modules with sensor logic to accurately locate and report structural faults in railway tracks.
- Enhanced railway safety automation by building a compact, cost-effective system for track monitoring, reducing dependency on manual inspection.

Education

University of Michigan, Dearborn Sep 2024 – May 2026

M.S. in Computer and Information Science, GPA: 3.94/4.0 *Dearborn, MI*

Relevant Coursework: Computer Networks, Advanced Operating Systems, Cloud Computing

Sri Venkateswara College of Engineering, Anna University Aug 2020 – May 2024

B.E. in Electronics and Communication, GPA: 8.23/10 *Chennai, India*

Relevant Coursework: Microcontrollers, IoT Systems, Object-Oriented Programming, Data Structures

Technical Skills

Languages: Python, Java, C, C++, JavaScript, SQL
Web & Backend: HTML, CSS, React.js, Node.js, Express.js, Django, Flask, Spring Boot, Spring Framework, Hibernate, JSP, Servlets, JDBC, SOAP, RESTful API Design, JSON, Client–Server Architecture, Microservices, Web Services
Databases: MySQL, MongoDB, DynamoDB, SQL
Cloud Computing (AWS): EC2, S3, Lambda, API Gateway, DynamoDB, IAM, SNS, SQS, CloudWatch, Rekognition, Kinesis Video Streams, Serverless Architecture, Event-Driven Systems, Fault Tolerance, Scalability, High Availability; also familiar with Google Cloud Platform (GCP)
Networking & Operating Systems: TCP/IP, HTTP/HTTPS, DNS, DHCP, NAT, IP Addressing, Subnetting, CIDR, Routing Tables, VPC Architecture, Subnets, Security Groups, Load Balancing, Linux, macOS, Windows, Network Security, Encryption
AI, ML & Deep Learning: Supervised/Unsupervised Learning, Feature Engineering, Model Training & Evaluation, Bias–Variance Tradeoff, Cross-Validation, Neural Networks, CNNs, RNNs, LSTMs, Bi-LSTMs, Backpropagation, TensorFlow, Keras, Computer Vision, Information Retrieval, Data Science
Embedded Systems & Hardware: Arduino IDE, Microcontrollers (8051, ESP32), IoT, Sensors, GSM, GPS, PLC, SCADA, Embedded C, IAR Embedded Workbench, Cadence Software, Tanner EDA, Xilinx ISE, VLSI
DevOps & Tools: Git, Jenkins, Terraform, Ansible, Docker, Kubernetes, Maven, Gradle, JUnit, Playwright, Postman, Cisco Packet Tracer, Mininet
Platforms & Enterprise: ServiceNow (ITOM, CMDB, ITSM), MATLAB, Power BI, Streamlit, PyCharm
Core CS: Data Structures & Algorithms, Object-Oriented Design, System Design, Full-Stack Development

Certifications & Courses

Programming in Java – NPTEL Jun 2023 – Oct 2023
Programming in Python – NPTEL Jan 2023 – Apr 2023